# Chapter 7

(Week 13)

# The Application Layer

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# PREVIOUS LAYERS

- THE PURPOSE OF THE PHYSICAL LAYER IS TO TRANSPORT A RAW BIT STREAM FROM ONE MACHINE TO ANOTHER.
- THE MAIN TASK OF THE DATA LINK LAYER IS TO TRANSFORM A RAW TRANSMISSION FACILITY INTO A LINE THAT APPEARS FREE OF UNDETECTED TRANSMISSION ERRORS TO THE NETWORK LAYER.

- THE NETWORK LAYER IS CONCERNED WITH GETTING PACKETS FROM THE SOURCE ALL THE WAY TO THE DESTINATION.
- THE TRANSPORT LAYER'S TASK IS TO PROVIDE RELIABLE, COST-EFFECTIVE DATA TRANSPORT FROM SOURCE MACHINE TO DESTINATION MACHINE, INDEPENDENTLY OF THE PHYSICAL NETWORK OR NERWORKS CURRENTLY IN USE.

# The Application Layer

- Layers below the application layer are there to provide reliable transport, but they do not do real work for users.

- In this chapter we will study some real network applications.

# The Application Layer

- In the application layer there is also need for support protocols, to allow the applications to function.
- One of them is DNS which handles naming within the Internet.
- Three real applications:

1. Electronic Mail
2. The World Wide Web
3. Multimedia

# The Application Layer's topics

7.1. DNS – The Domain Name System

7.2. Electronic Mail

7.3. The World Wide Web

7.4. Multimedia

7.5. Summary

# 7.1. DNS – The Domain Name System

- 7.1.1. The DNS Name Space
- 7.1.2. Resource Records
- 7.1.3. Name Servers

# 7.1. DNS – The Domain Name System

- Network addresses (e.g., IP) are hard for people to remember.
- Also, sending e-mail to tana@128.111.24.41 means that if Tana's ISP moves the mail server to a different machine with different IP address, her e-mail address has to change.

# 7.1. DNS – The Domain Name System

- Consequently, ASCII names were introduced to decouple machine names from machine addresses.
- In this way, Tana's address might be something like tana@art.ucsb.edu.
- Nevertheless, network itself understand only numerical addresses, so some mechanism is required to convert the ASCII srtings to network addresses.

# 7.1. DNS – The Domain Name System

- DNS (Domain Name System) solves this problem.
- The essence of DNS is the invention of a hierarchical, domain-based naming scheme and a distributed database system for implementing this naming scheme.
- DNS is primarily used for mapping host names and e-mail destinations to IP addresses but can also be used for other purposes.

# 7.1. DNS – The Domain Name System

- DNS is used as follows.
- To map a name onto an IP address, an application program calls a library procedure called the resolver, passing it the name as a parameter.

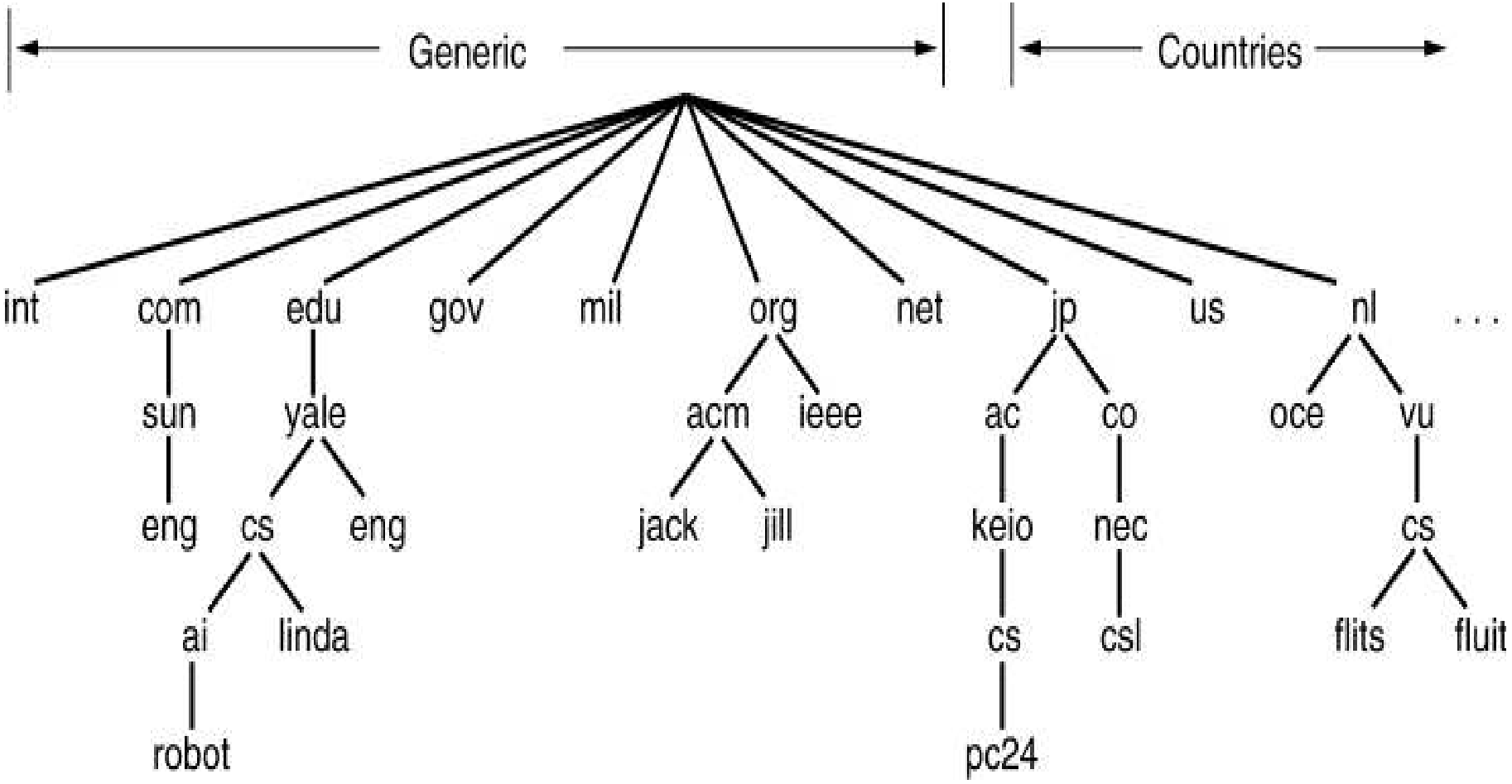
# 7.1. DNS – The Domain Name System

- Resolver sends a UDP packet to a local DNS server, which then looks up the name and returns the IP address to the resolver, which then returns it to the caller.
- Armed with the IP address, the program can then establish a TCP connection with the destination or send it UDP packets.

# 7.1.1. The DNS Name Space

- Internet is divided into over 200 top-level domains, where each domain covers many hosts.
- Each domain is partitioned into subdomains, and these are further partitioned, and so on.
- All these domains can be represented by a tree.

# 7.1.1. The DNS Name Space




A portion of the Internet domain name space.

# 7.1.1. The DNS Name Space

- The leaves of the tree represent domains that have no subdomains (but do contain machines, of course).
- A leaf domain may contain a single host, or it may represent a company and contain thousands of hosts.
- The top-level domains come in two flavors: generic and countries.

# 7.1.1. The DNS Name Space

- Top-level general-purpose domains:
- Commercial (com)
- Educational institutions (edu)
- Governments (gov)
- Certain international organizations (int)
- Armed forces (mil)
- Network providers (net)
- Nonprofit organizations (org)

# 7.1.1. The DNS Name Space

- Business (biz) – 2000, ICANN
- Information (inf) – 2000, ICANN
- Peaple's names (name) – 2000, ICANN
- Professions (pro) – 2000, ICANN
- Aerospace industry (aero) - by request
- Co-operatives (coop) - by request
- Museums (museum) - by request, etc

# 7.1.1. The DNS Name Space

- In general, getting a second-level domain, such as name-of-company.com, is easy.
- It requires going to a registrar for the corresponding top-level domain (com in this case) to check if the desired name is available and not somebody's trademark.
- If there are no problems, the register pays a small annual fee and gets the name.

# 7.1.1. The DNS Name Space

- Each domain is named by the path upward from it to the root.
- The components are separated by periods (pronounced “dot”).
- *eng.sun.com* is the engineering department at Sun Microsystems.
- *eng/sun/com* is UNIX-style

# 7.1.1. The DNS Name Space

- Domain names are case insensitive, so edu, Edu, and EDU mean the same thing.
- Component names can be up to 63 characters long, and full path names must not exceed 255 characters.
-

# 7.1.1. The DNS Name Space

- Domains can be inserted into the tree in two different ways.
- *cs.yale.edu*
- *cs.yale.ct.us*
- Most organization in the United States are under a generic domain, and most outside the United States are under the domain of their country.

# 7.1.1. The DNS Name Space

- Each domain controls how it allocates the domains under it.
- For example, Japon has domains ac.jp and co.jp that mirror edu and com
- The Netherlands does not make this distinction and puts all organization directly under nl

# 7.1.1. The DNS Name Space

- Thus, all three of the following are university computer science departments:
- cs.yale.edu (Yale University, in the US)
- cs.vu.nl (Vrije Universiteit, in the Netherl)
- cs.keio.ac.jp (Keio University, in Japan)
- Naming follows organizational boundaries, not physical networks.

# 7.1.2. Resource Records

- Every domain, whether it is a single host or a top-level domain, can have a set of resource records associated with it.
- For a single host, the most common resource record is just its IP address, but many other kinds of resource records also exist.

# 7.1.2. Resource Records

- When a resolver gives a domain name to DNS, what it gets back are the resource records associated with that name.
- Thus, the primary function of DNS is to map domain names onto resource records.

# 7.1.2. Resource Records

- A resource record is a five-tuple.
- Resource records are presented as ASCII text, one line per resource record.
- The format we will use is as follows:
- Domain_name Time_to_live Class Type Value

# 7.1.2. Resource Records

- Domain_name
- It tells the domain to which this record applies.
- Normally, many records exist for each domain and each copy of the database holds information about multiple domains.
- This field is thus the primary search key used to satisfy queries.

# 7.1.2. Resource Records

- Time_to_live
- It gives an indication of how stable the record is.
- Information that is highly stable is assigned a large value, such as 86400 (the number of seconds in 1 day).
- Information that is highly volatile is assigned a small value, such as 60 (1 minute).

# 7.1.2. Resource Records

- Class
- For Internet information, it is always *IN*.
- Type
- It tells what kind of record this is
- The most important types are listed in following figure.

# 7.1.2. Resource Records

| Type | Meaning | Value |
|---|---|---|
| SOA | Start of Authority | Parameters for this zone |
| A | IP address of a host | 32-Bit integer |
| MX | Mail exchange | Priority, domain willing to accept e-mail |
| NS | Name Server | Name of a server for this domain |
| CNAME | Canonical name | Domain name |
| PTR | Pointer | Alias for an IP address |
| HINFO | Host description | CPU and OS in ASCII |
| TXT | Text | Uninterpreted ASCII text |

The principal DNS resource records types.

# 7.1.2. Resource Records

- Value
- This field can be a number, a domain name, or an ASCII string.
- The semantics depend on the record type.
- Following figure depicts part of a database for the cs.vu.nl domain shown on slide number 12.

# 7.1.2. Resource Records

```
; Authoritative data for cs.vu.nl
cs.vu.nl.          86400   IN   SOA      star boss (952771,7200,7200,2419200,86400)
cs.vu.nl.          86400   IN   TXT      "Divisie Wiskunde en Informatica."
cs.vu.nl.          86400   IN   TXT      "Vrije Universiteit Amsterdam."
cs.vu.nl.          86400   IN   MX       1 zephyr.cs.vu.nl.
cs.vu.nl.          86400   IN   MX       2 top.cs.vu.nl.

flits.cs.vu.nl.    86400   IN   HINFO    Sun Unix
flits.cs.vu.nl.    86400   IN   A        130.37.16.112
flits.cs.vu.nl.    86400   IN   A        192.31.231.165
flits.cs.vu.nl.    86400   IN   MX       1 flits.cs.vu.nl.
flits.cs.vu.nl.    86400   IN   MX       2 zephyr.cs.vu.nl.
flits.cs.vu.nl.    86400   IN   MX       3 top.cs.vu.nl.
www.cs.vu.nl.      86400   IN   CNAME    star.cs.vu.nl
ftp.cs.vu.nl.      86400   IN   CNAME    zephyr.cs.vu.nl

rowboat                    IN   A        130.37.56.201
                           IN   MX       1 rowboat
                           IN   MX       2 zephyr
                           IN   HINFO    Sun Unix

little-sister              IN   A        130.37.62.23
                           IN   HINFO    Mac MacOS

laserjet                   IN   A        192.31.231.216
                           IN   HINFO    "HP Laserjet IIISi" Proprietary
```

A portion of a possible DNS database for *cs.vu.nl.*

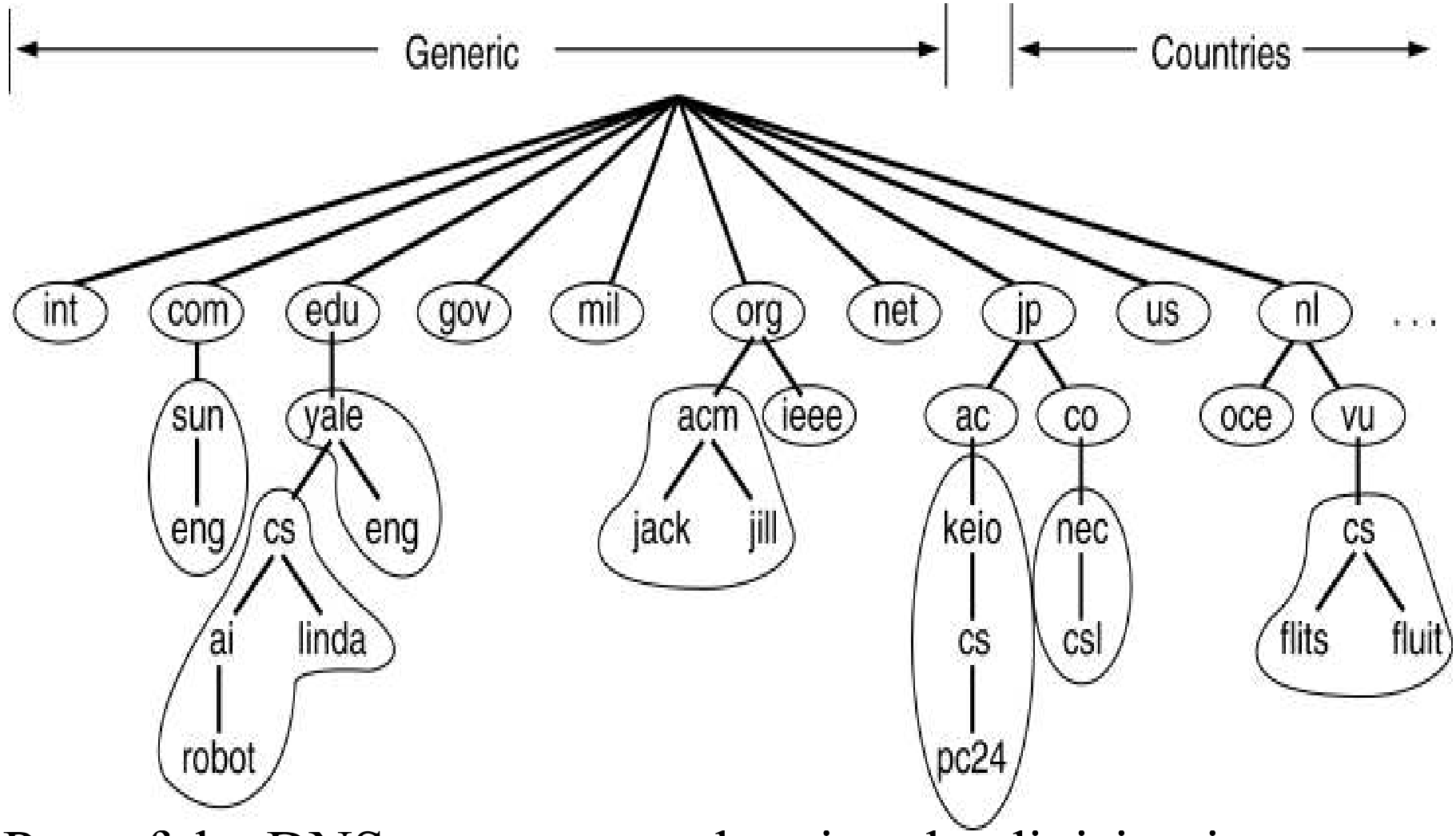
# 7.1.3. Name Servers

- In theoty at least, a single name server could contain the entire DNS database and respond to all queries about it.
- In practice, this server would be so overloaded as to be useless.
- Futhermore, if it ever went down, the entire Internet would be crippled.

# 7.1.3. Name Servers

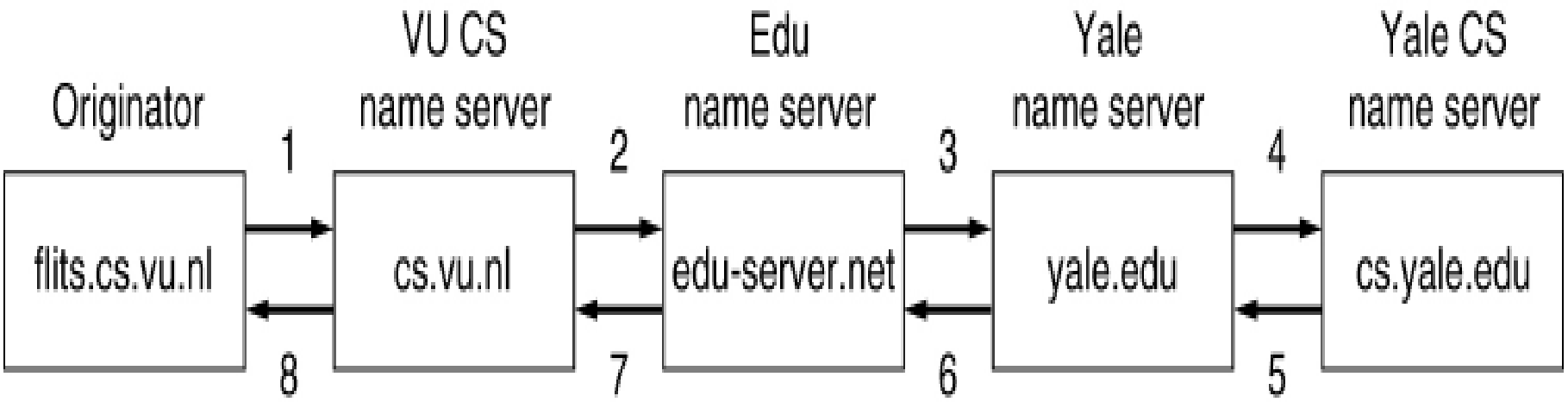
- To avoid the problems associated with having only a single source of information, the DNS name space is divided into nonoverlapping zones.
- One possible way to divide the name space on slide 12 is shown in following figure.

# 7.1.3. Name Servers




Part of the DNS name space showing the division into zones.

# 7.1.3. Name Servers




How a resolver looks up a remote name in eight steps.

# 7.2. Electronic Mail

- Electronic mail, or e-mail, like other forms of communication, has its own conventions and style.
- e-mail is very informal and has a low threshold of use.
- e-mail is full of jargon
- Many people also use litle ASCII symbols called smileys or emoticons in their e-mail…

# 7.2. Electronic Mail

| Smiley | Meaning | Smiley | Meaning | Smiley | Meaning |
|---|---|---|---|---|---|
| :-) | I'm happy | =\|:-) | Abe Lincoln | :+) | Big nose |
| :-( | I'm sad/angry | =):-) | Uncle Sam | :-)) | Double chin |
| :-\| | I'm apathetic | *<:-) | Santa Claus | :-{) | Mustache |
| ;-) | I'm winking | <:-( | Dunce | #:-) | Matted hair |
| :-(O) | I'm yelling | (-: | Australian | 8-) | Wears glasses |
| :-(*) | I'm vomiting | :-)X | Man with bowtie | C:-) | Large brain |

Some smileys. They will not be on the final exam :-).

# 7.2. Electronic Mail

- The first e-mail systems simply consisted of file transfer protocols, with the convention that the first line of each message (i.e., file) contained the recipient's address.

# 7.2. Electronic Mail

- Some of the complaints were as follows:

a) Sending a message to a group of people was inconvenient.

b) Messages had no internal structure, making computer processing difficult.

c) The sender never knew if a message arrived or not.

d) It was not possible to create and send messages containing a mixture of text, drawings, facsimile, and voice, etc.

# 7.2. Electronic Mail

- 1982 – RFC 821 (transmission protocol) and RFC 822 (message format)
- 1984 – X.400
- After two decades of competition, e-mail systems based on RFC 822 are widely used.

# 7.2. Electronic Mail

- Architecture and Services
- The User Agent
- Message Formats
- Message Transfer
- Final Delivery

# 7.2.1. Architecture and Services

- What e-mail systems can do and how they are organized?
- There are two subsystems in e-mail systems

a) User agents allow people to read and send e-mail.

b) Message transfer agents move messages from the source to the destination.

# 7.2.1. Architecture and Services

e-mail systems support five basic functions

- Composition
- Transfer
- Reporting
- Displaying
- Disposition

# 7.2.1. Architecture and Services

A few of advanced features:

- Mailboxes
- Mailing list
- Carbon copies
- Blind carbon copies
- Etc.

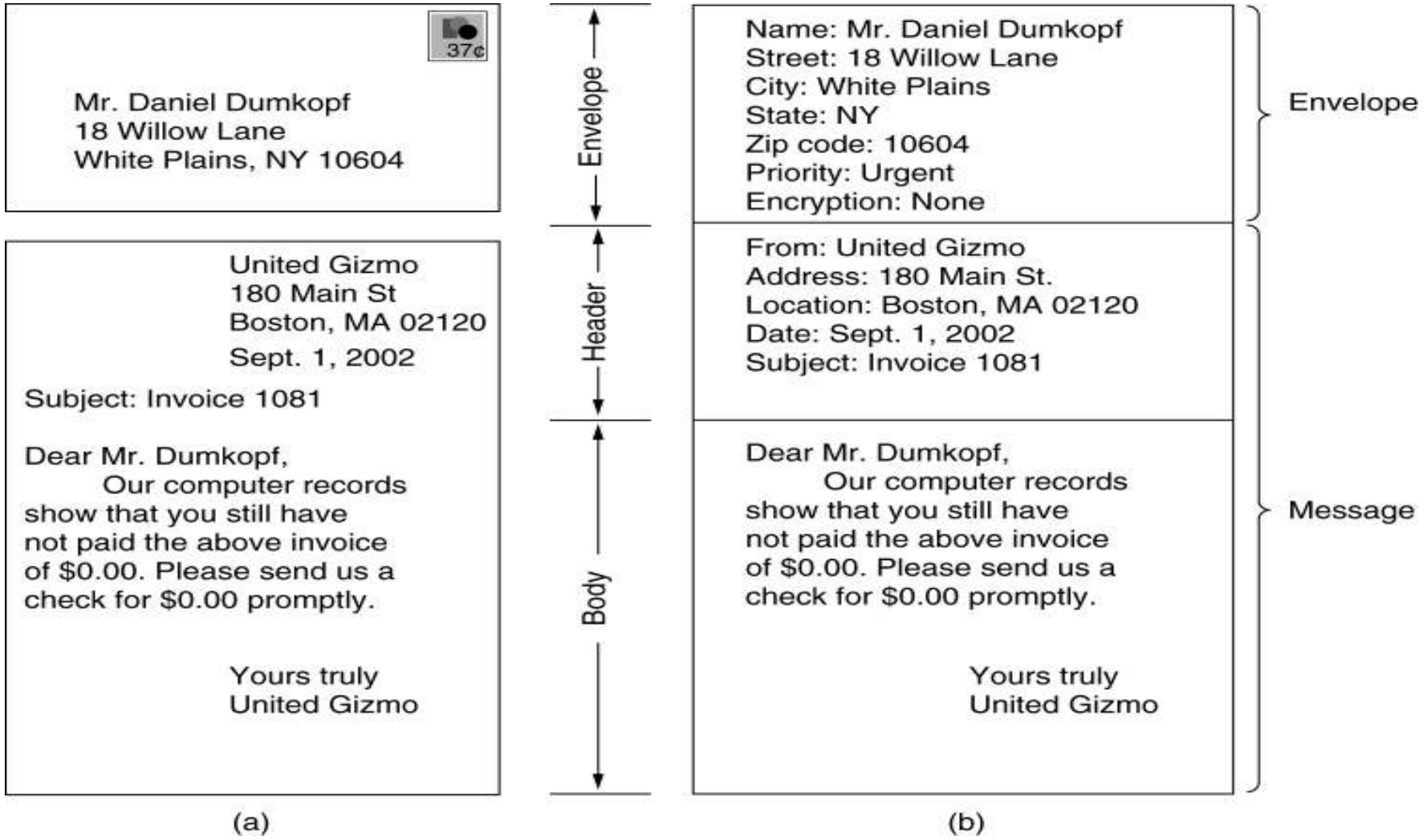
# 7.2.1. Architecture and Services

- A key idea in e-mail systems is the distinction between the envelope and its contents.
- The envelope encapsulates the message
- It contains all the information needed for transporting the message, such as the destination address, priority, and security level, all of which are distinct from the message itself.

# 7.2.1. Architecture and Services

- Message inside the envelope consists of two parts:
- The header and the body
- The header contains control information for the user agents.
- The body is entirely for the human recipient.

# 7.2.1. Architecture and Services




Envelopes and messages. (a) Paper mail. (b) Electronic mail.

# 7.2.2. The User Agent

- A user agent is normally a program (sometimes called a mail reader) that accepts a variety of commands for composing, receiving, and replying to message, as well as for manipulating mailboxes.

# 7.2.2. The User Agent

- SENDING E-MAIL
- To send an e-mail message, a user must provide the message, the destination address, and possiply some other parameters,
- The message can be produced with a free-standing text editor, a word processing program, etc
- Most e-mail systems support mailing lists, so that a user can send same message to a list of people with a single command.

# 7.2.2. The User Agent

- READING E-MAIL
- Typically, when a user agent is started up, it looks at the user's mailbox for incoming e-mail before displaying anything on the screen.
- Then it may announce the number of messages in the mailbox or display a one-line summary of each one and wait for a command.

# 7.2.2. Reading E-mail

| # | Flags | Bytes | Sender | Subject |
|---|---|---|---|---|
| 1 | K | 1030 | asw | Changes to MINIX |
| 2 | KA | 6348 | trudy | Not all Trudys are nasty |
| 3 | K F | 4519 | Amy N. Wong | Request for information |
| 4 | | 1236 | bal | Bioinformatics |
| 5 | | 104110 | kaashoek | Material on peer-to-peer |
| 6 | | 1223 | Frank | Re: Will you review a grant proposal |
| 7 | | 3110 | guido | Our paper has been accepted |
| 8 | | 1204 | dmr | Re: My student's visit |

An example display of the contents of a mailbox.

# 7.2.3 Message Formats – RFC 822

- Messages consist of :
- a primitive envelope,
- some number of header fields,
- a blank line,
- and then message body

# 7.2.3 Message Formats – RFC 822

- Each header field consists of a single line of ASCII text containing the field name, a colon, and, for most fields, a value.

- In normal usage, the user agent builds a message and passes it to the message transfer agent, which then uses some of the header fields to construct the actual envelope.

# 7.2.3 Message Formats – RFC 822

| Header | Meaning |
|---|---|
| To: | E-mail address(es) of primary recipient(s) |
| Cc: | E-mail address(es) of secondary recipient(s) |
| Bcc: | E-mail address(es) for blind carbon copies |
| From: | Person or people who created the message |
| Sender: | E-mail address of the actual sender |
| Received: | Line added by each transfer agent along the route |
| Return-Path: | Can be used to identify a path back to the sender |

RFC 822 header fields related to message transport.

# Message Formats – RFC 822 (2)

| Header | Meaning |
|---|---|
| Date: | The date and time the message was sent |
| Reply-To: | E-mail address to which replies should be sent |
| Message-Id: | Unique number for referencing this message later |
| In-Reply-To: | Message-Id of the message to which this is a reply |
| References: | Other relevant Message-Ids |
| Keywords: | User-chosen keywords |
| Subject: | Short summary of the message for the one-line display |

Some fields used in the RFC 822 message header.

# MIME – Multipurpose Internet Mail Extensions

- In the early days of the ARPANET, e-mail consisted exclusivly of text messages written in English and expressed in ASCII.
- For this environment, RFC 822 did the job completely: it specified the headers but left the content entirely up to the users.

# MIME – Multipurpose Internet Mail Extensions

Problems with international languages:

- Languages with accents (French, German).
- Languages in non-Latin alphabets (Hebrew, Russian).
- Languages without alphabets (Chinese, Japanese).
- Messages not containing text at all (audio or images).

# MIME – Multipurpose Internet Mail Extensions

- A solution was proposed in RFC 1341 and updated in RFCs 2045-2049.
- This solution, called MIME (Multipurpose Internet Mail Extensions) is now widely used.
- The basic idea of MIME is to continue to use the RFC 822 format, but to add srtucture to the message body and define encoding rules for non-ASCII messages.

# MIME (2)

| Header | Meaning |
|---|---|
| MIME-Version: | Identifies the MIME version |
| Content-Description: | Human-readable string telling what is in the message |
| Content-Id: | Unique identifier |
| Content-Transfer-Encoding: | How the body is wrapped for transmission |
| Content-Type: | Type and format of the content |

RFC 822 headers added by MIME.

# MIME (3)

| Type | Subtype | Description |
|---|---|---|
| Text | Plain | Unformatted text |
| | Enriched | Text including simple formatting commands |
| Image | Gif | Still picture in GIF format |
| | Jpeg | Still picture in JPEG format |
| Audio | Basic | Audible sound |
| Video | Mpeg | Movie in MPEG format |
| Application | Octet-stream | An uninterpreted byte sequence |
| | Postscript | A printable document in PostScript |
| Message | Rfc822 | A MIME RFC 822 message |
| | Partial | Message has been split for transmission |
| | External-body | Message itself must be fetched over the net |
| Multipart | Mixed | Independent parts in the specified order |
| | Alternative | Same message in different formats |
| | Parallel | Parts must be viewed simultaneously |
| | Digest | Each part is a complete RFC 822 message |

The MIME types and subtypes defined in RFC 2045.

# MIME (4)

```
From: elinor@abcd.com
To: carolyn@xyz.com
MIME-Version: 1.0
Message-Id: <0704760941.AA00747@abcd.com>
Content-Type: multipart/alternative; boundary=qwertyuiopasdfghjklzxcvbnm
Subject: Earth orbits sun integral number of times

This is the preamble. The user agent ignores it. Have a nice day.

--qwertyuiopasdfghjklzxcvbnm
Content-Type: text/enriched

Happy birthday to you
Happy birthday to you
Happy birthday dear <bold> Carolyn </bold>
Happy birthday to you

--qwertyuiopasdfghjklzxcvbnm
Content-Type: message/external-body;
        access-type="anon-ftp";
        site="bicycle.abcd.com";
        directory="pub";
        name="birthday.snd"

content-type: audio/basic
content-transfer-encoding: base64
--qwertyuiopasdfghjklzxcvbnm--
```

A multipart message containing enriched and audio alternatives.

# 7.2.4. Message Transfer

- The message transfer system is concerned with relaying messages from the originator to the recipient.

- The simplest way to do this is to establish a transport connection from the source machine to the destination machine and then just transfer the message.

# 7.2.4. Message Transfer

- SMTP – The Simple Mail Transfer Protocol
- Within the Internet, e-mail is delivered by having the source machine establish a TCP connection to port 25 of the destination machine.
- Listening to this port is an e-mail daemon that speaks SMTP.

# 7.2.4. Message Transfer

- This daemon accepts incoming connections and copies messages from them into the appropriate mailboxes.

- If a message cannot be delivered, an error report containing the first part of the undeliverable message is returned to the sender.

# 7.2.4. Message Transfer

- SMTP is a simple ASCII protocol.

- After establishing the TCP connection to port 25, the sending machine, operating as the client, waits for the receiving machine, operating as the server, to talk first.

# 7.2.4. Message Transfer

- The server starts by sending a line of text giving its identity and telling whether it is prepared to receive mail.

- If it is not, the client releases the connection and tries again later.

# 7.2.4. Message Transfer

- If the server is willing to accept e-mail, the client announces whom the e-mail is coming from and whom it is going to.

- If such a recipient exists at the destination, the server gives the client the go-ahead to send the message.

# 7.2.4. Message Transfer

- Then the client sends the message and the server acknowledges it.
- No checksums are needed because TCP provides a reliable byte stream.
- If there is more e-mail, that is now sent.
- When all the e-mail has been exchanged in both directions, the connection is released.

# Message Transfer

Transferring a message from *elinore@abc.com* to *carolyn@xyz.com*.

```
                    S: 220 xyz.com SMTP service ready
C: HELO abcd.com
                    S: 250 xyz.com says hello to abcd.com
C: MAIL FROM: <elinor@abcd.com>
                    S: 250 sender ok
C: RCPT TO: <carolyn@xyz.com>
                    S: 250 recipient ok
C: DATA
                    S: 354 Send mail; end with "." on a line by itself
C: From: elinor@abcd.com
C: To: carolyn@xyz.com
C: MIME-Version: 1.0
C: Message-Id: <0704760941.AA00747@abcd.com>
C: Content-Type: multipart/alternative; boundary=qwertyuiopasdfghjklzxcvbnm
C: Subject: Earth orbits sun integral number of times
C:
C: This is the preamble. The user agent ignores it. Have a nice day.
C:
C: --qwertyuiopasdfghjklzxcvbnm
C: Content-Type: text/enriched
C:
C: Happy birthday to you
C: Happy birthday to you
C: Happy birthday dear <bold> Carolyn </bold>
C: Happy birthday to you
C:
C: --qwertyuiopasdfghjklzxcvbnm
C: Content-Type: message/external-body;
C:          access-type="anon-ftp";
C:          site="bicycle.abcd.com";
C:          directory="pub";
C:          name="birthday.snd"
C:
C: content-type: audio/basic
C: content-transfer-encoding: base64
C: --qwertyuiopasdfghjklzxcvbnm
C: .
                    S: 250 message accepted
C: QUIT
                    S: 221 xyz.com closing connection
```

# 7.2.5. Final Delivery

- What happens when Elinor wants to send Carolyn e-mail and Carolyn is not currently on-line?
- Elinor cannot establish a TCP connection to Carolyn and thus cannot run the SMTP protocol.

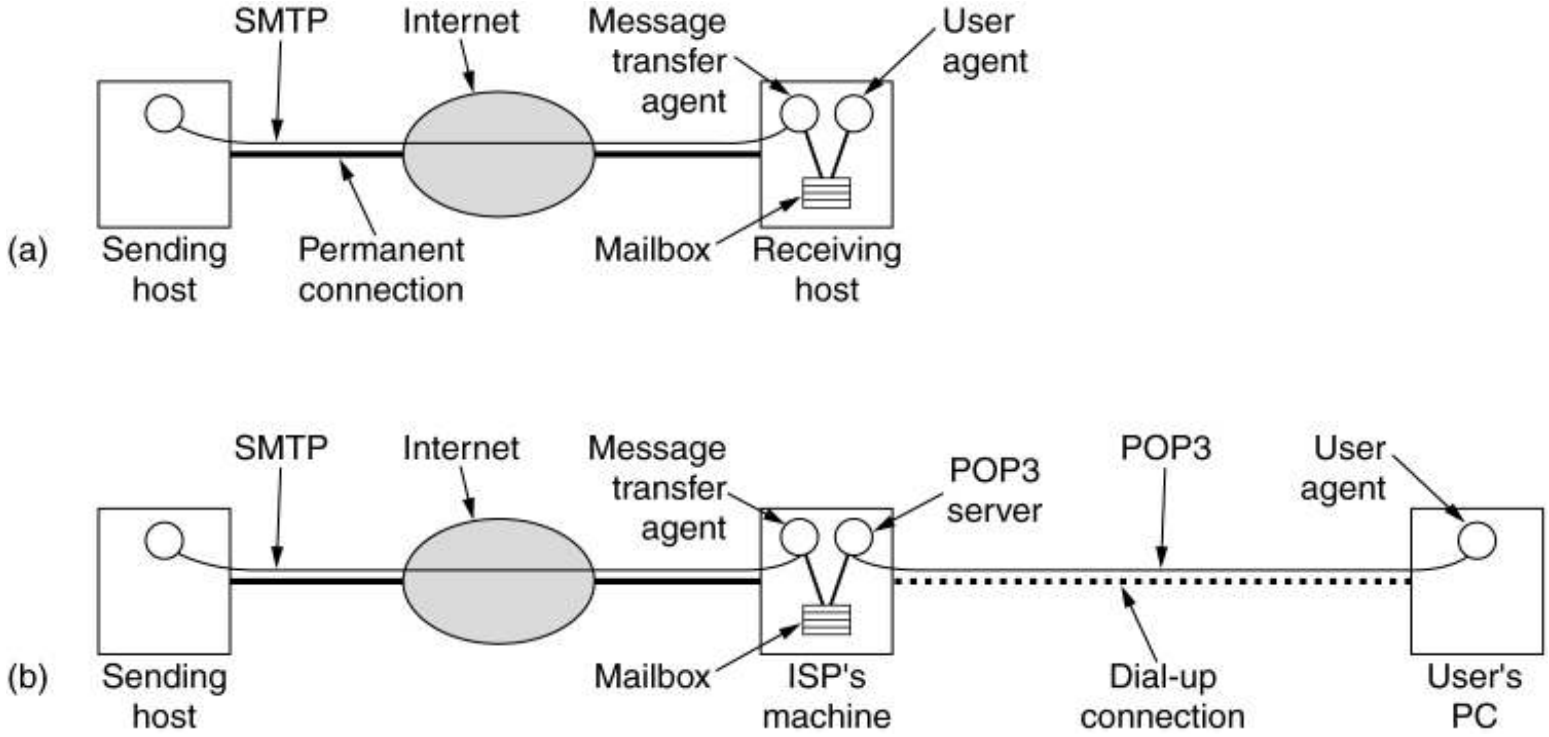
# 7.2.5. Final Delivery

- One solution is to have a message transfer agent on an ISP machine accept e-mail for its customers and store it in their mailboxes on an ISP machine.
- Since this agent can be on-line all the time, e-mail can be sent to it 24 hours a day.

# 7.2.5. Final Delivery - POP3

- How does the user get the e-mail from the ISP's message transfer agent?
- The solution to this problem is to create another protocol that allows user transfer agents (on client PCs) to contact the message transfer agent (on ISP's machine) and allow e-mail to be copied from the ISP to the user.
- POP3 – Post Office Protocol Version 3

# 7.2.5. Final Delivery




(a) Sending and reading mail when the receiver has a permanent Internet connection and the user agent runs on the same machine as the message transfer agent. (b) Reading e-mail when the receiver has a dial-up connection to an ISP.

# 7.2.5. Final Delivery - POP3

```
                    S: +OK POP3 server ready
C: USER carolyn
                    S: +OK
C: PASS vegetables
                    S: +OK login successful
C: LIST
                    S: 1 2505
                    S: 2 14302
                    S: 3 8122
                    S: .
C: RETR 1
                    S: (sends message 1)
C: DELE 1
C: RETR 2
                    S: (sends message 2)
C: DELE 2
C: RETR 3
                    S: (sends message 3)
C: DELE 3
C: QUIT
                    S: +OK POP3 server disconnecting
```

Using POP3 to fetch three messages.

# 7.2.5. Final Delivery - IMAP

- IMAP – Internet Message Access Protocol
- IMAP assumes that all the e-mail will remain on the server indefinitely in multiple mailboxes.
- But POP3 basically assumes that the user will clear out the mailbox on every contact and work off-line after that.
- IMAP – provides extensive mechanisms for reading message or even parts of messages.

# IMAP

| Feature | POP3 | IMAP |
|---|---|---|
| Where is protocol defined? | RFC 1939 | RFC 2060 |
| Which TCP port is used? | 110 | 143 |
| Where is e-mail stored? | User's PC | Server |
| Where is e-mail read? | Off-line | On-line |
| Connect time required? | Little | Much |
| Use of server resources? | Minimal | Extensive |
| Multiple mailboxes? | No | Yes |
| Who backs up mailboxes? | User | ISP |
| Good for mobile users? | No | Yes |
| User control over downloading? | Little | Great |
| Partial message downloads? | No | Yes |
| Are disk quotas a problem? | No | Could be in time |
| Simple to implement? | Yes | No |
| Widespread support? | Yes | Growing |

A comparison of POP3 and IMAP.